

PROBLEM SET 9

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 10)

1) (10 pts) A particle is in a potential

$$V(x) = V_0 \sin(2\pi x/a)$$

which is invariant under the translations $x \rightarrow x + ma$, where m is an integer. Verify whether or not momentum is conserved.

If momentum is not, then why not?

Exercise 11.4.2 (p. 300)

$$\hat{H} = \frac{\hat{P}^2}{2m} + V_0 \sin\left(\frac{2\pi}{a} \hat{X}\right).$$

For momentum conservation to hold,

$$\frac{d}{dt} \langle \hat{P} \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{P}] \rangle = 0 \quad \forall \text{ states}$$

we need $[\hat{H}, \hat{P}] = 0$.

However, here

$$[\hat{H}, \hat{P}] = \left[\frac{\hat{P}^2}{2m} + V_0 \sin\left(\frac{2\pi}{a} \hat{X}\right), \hat{P} \right] = V_0 \left[\sin\left(\frac{2\pi}{a} \hat{X}\right), \hat{P} \right] \neq 0$$